GEOPHYSICAL RESEARCH LETTERS, VOL. 19, NO. 21, PAGES 2147-2150, NOVEMBER 3, 1992

THE PALEOMAGNETIC RECORD OF UPPERMOST PERMIAN, LOWER TRIASSIC ROCKS FROM THE SOUTH CHINA BLOCK

R.J. Enkin[1], V. Courtillot[2], Ph. Leloup[2], Z. Yang[2], L. Xing[3], J. Zhang[3], and Z. Zhuang[3]

Abstract. In order to constrain better the paleogeography of the blocks that now make up China, we collected paleomagnetic samples from Uppermost Permian to Middle Triassic limestones from two localities in the Sichuan Basin (South China Block). A low inclination component which passes both the fold test and the reversal test was found in five sites (D=46.1°, I=11.9°, k=83.6, α_{95}=8.7°). We review corresponding remanences found in other studies of Permo-Triassic rocks from the Sichuan Basin, and show that the component is probably characteristic of the Lower Triassic. We then review paleomagnetic results from the Uppermost Permian Emeishan Basalt. Despite small local rotations, the difference between the Emeishan Basalts and the Sichuan Basin poles is interpreted to be the result of motion of the whole South China Block around the time of the Permo-Triassic boundary rather than of strain within the block.

Introduction

The cratonic blocks of China and south-east Asia have had a complex history of separation from Gondwanaland, accretion onto the Eurasian core and deformation due to the subsequent accretion of other blocks, notably India [e.g., *Zhang et al.*, 1984, *Tapponnier et al.*, 1986]. With the goal of better defining the age of accretion of the South China Block (SCB) and the North China Block and to constrain further the effects of the subsequent deformation, we collected oriented samples of Upper Permian to Middle Triassic Limestones from the Sichuan Basin (SCB).

Geology and Fieldwork

The Permian and Triassic stratigraphy of the SCB is dominated by white limestones. Two important departures from this relatively monotonous sequence are the late Permian Emeishan Basalts and the early Triassic Feixienguan Formation with its distinctive red-purple limestones and mudstones [*Yang et al.*, 1986]. Both of these formations are excellent carriers of magnetic remanence.

The Upper Permian Emeishan Basalts cover a wide area of south-west Sichuan and continue into neighbouring Yunnan and Guizhou provinces (Figure 1). The flows are overlain by the uppermost Permian Xuanwei Formation which is considered to be coeval with the widespread Changxing (P_{2c}) Formation [*Yang et al.*, 1986, p.117]. The conformable transition from these white limestone formations to the shallow marine or tidal flat Feixienguan Formation (T_{1f}) is taken to be the Permo-Triassic boundary. The stratigraphic column continues, in the Sichuan Basin, with the Lower Triassic Jialingjiang Formation (T_{1j}) and the Middle Triassic Leikoupo Formation (T_{2l}). The sea receded during the Middle to Upper Triassic, and continental clastic sediments only were deposited above the T_{2l} Formation. The bulk of the Sichuan basin is composed of Jurassic to Tertiary red-sandstones. Older marine facies outcrop only on the basin periphery and in the cores of long and narrow box-shaped anticlines separated by large flat synclines (Figure 1).

We collected Permian and Triassic marine facies rocks from two regions in the Sichuan Basin: an anticline north of Nanxi (28.9°N, 104.9°E, T1.S.12 in Figure 1, sites 77 to 82), and two sections north and west of Changshou (29.9°N, 107.0°E, T1.S.13 in Figure 1, sites 88 to 94, 104, 105). Standard 2.5 cm cores (3 to 10 cm long) were taken with a portable gas-powered drill, and oriented with a magnetic compass (the weather precluded solar orientations). Specimens, 2.2 cm in length, were cut from the cores in the laboratory.

Paleomagnetic Results

The samples were split evenly between the Beijing and Paris laboratories. Measurements were done with a

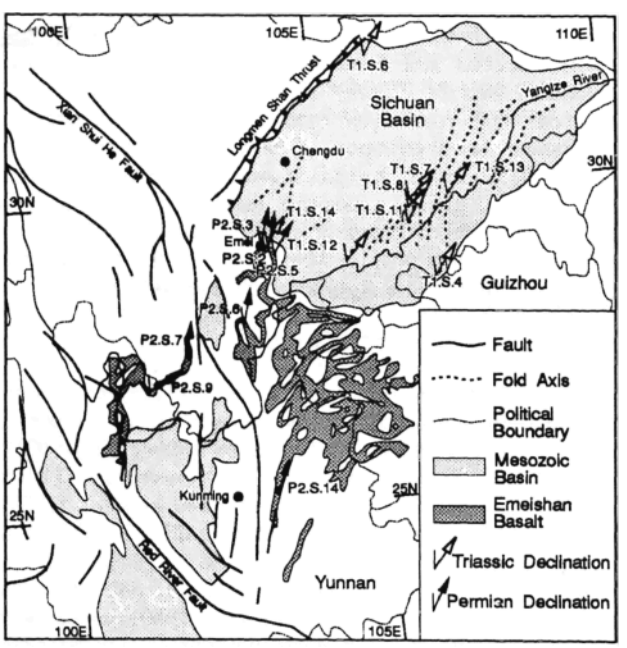


Fig. 1. Simplified tectonic map of Sichuan and surrounding region. Outside the Sichuan Basin, the main faults are shown, while within the basin the fold axes are shown. The declinations of Lower Triassic and Upper Permian Emeishan Basalt studies are identified with codes from *Enkin et al.* [1992] and given in the text. The present work includes data from Nanxi (T1.S.12) and Changshou (T1.S.13).

[1] Geology and Geophysics, University of Edinburgh, Scotland
[2] Institut de Physique du Globe de Paris, France
[3] Chinese Academy of Geological Sciences, Beijing, China

Copyright 1992 by the American Geophysical Union.

Paper number 92GL02321
0094-8534/92/92GL-02321$03.00

Schonstedt fluxgate magnetometer in Beijing, and with a CTF SQUID magnetometer in Paris. Thermal demagnetization was the most efficient method to separate components. At least 8 steps were measured for each sample.

We discovered three magnetic components in our collection. In almost all samples, we isolated a post-folding low-temperature component aligned with the present field.

In most sites from the T_{2l} and T_{1j} Formations we isolated a relatively steep normal-polarity direction. In site 92 (P_{2c}) a medium-temperature component also fits with this component. The directions cross over on tectonic correction, giving a very tight distribution at 50% unfolding (mean direction D=17.8°, I=52.1°, k=76.2, α_{95}=7.0°, n=7 sites). This direction must either be the unseparatable combination of a pre-folding Mesozoic overprint and the post-folding present-field overprint, or is the result of synfolding remanence (see *Enkin* [1990] for details).

From Upper Permian and Lower Triassic formations, we isolated a shallow inclination component with both polarities (Table 1, Figure 2, Figure 3). From the Nanxi section, only one site (79, T_{1f}) contained this component, but fortunately we collected a relatively large number of cores (15) from this site. Four sites from the Changshou area contain this component. Site 91 (T_{1j}) clearly contained an inverse south-west directed high temperature component, but only great circles (GCs) rather than final end points could be defined. Since the GCs are almost parallel we could define an average constrained GC by averaging the specimen GC normals and their sector constraints. Three linear segments could be isolated in most samples from site 92 (P_{2c}). The distribution of the site 92 high temperature component was slightly less well-clustered after tectonic correction, possibly due to contamination by the Mesozoic overprint.

The average direction of all these sites is, in stratigraphic coordinates, D=46.1°, I=11.9°, k=83.6, α_{95}=8.7°, for 4 site mean directions and 1 site mean great circle, using the *McFadden and McElhinny* [1988] method of combining directions and constrained great circles. In geographic coordinates the distribution is much more scattered, with mean D=37.6°, I=7.8°, k=13.0, α_{95}=22.6°. The two polarities are antipodal, giving a positive reversal test at the 95% significance level. According to *McElhinny*'s [1964] k-ratio test, the fold test is positive (k-ratio = 6.43 > 3.44, the 95% confidence limit for no significant change with N = 5).

Discussion

Similar directions, presumed to be primary, have been isolated from 7 localities in the Sichuan Basin (Figure 1, poles identified with the codes given by *Enkin et al.* [1992]): Tongzi T1.S.4 [*Opdyke et al.*, 1987], Guangyuan T1.S.6 [*Heller et al.*, 1988] (also sampled by *Steiner et al.* [1989] giving T1.S.5), two sections at Hechuan T1.S.7 and T1.S.8 [*Steiner et al.*, 1989], Liangfenya T1.S.11 [*Steiner et al.*, 1989], Nanxi T1.S.12 and Changshou T1.S.13 defined in this paper, and Emei T1.S.14 [*Ma and Zhang*, 1989]. Since most of these studies consist of magnetostratigraphic sections taken as a single site, mean directions have been reported at the sample rather than site level. Thus our results and that of *Opdyke et al.* [1987] are recalculated using sample instead of site averages. The sample level mean pole from Nanxi (T1.S.12) is 43.3°N, 211.7°E, dp=3.3, dm=6.4, and from Changshou (T1.S.13) 39.8°N, 216.2°E, dp=2.7, dm=5.3. The approximate sample level mean pole for Tongzi (T1.S.4, from site data given in *Opdyke et al.* [1987]) is 44.5°N, 217.8°E, dp=2.7°, dm=5.3°.

The Sichuan Basin poles have compatible paleolatitudes, all lying on a small circle of 84.9°±2.2° radius centered on Emei at 29.6°N, 103.4°E (Figure 4a, calculated using the method of *McFadden and Reid* [1982]). (Emei was chosen as an arbitrary central point, not significantly separated from any position in the Sichuan Basin - Emeishan Basalt region for the purpose of showing deviations in paleolatitude and paleoazimuth). There is, on the other hand, some dispersion in the declinations, which suggests that the sites have been subject to small (<15°) relative rotations. *Enkin* [1990] tested the hypothesis put forward by *Steiner et al.* [1989] that the the rotations are due to oroclinal bending, but found declinations were not well-correlated with the structural trends.

While the poles considered here come from formations dated as Upper Permian (P_{2c}) as well as Lower Triassic (T_{1f} and T_{1j}), observe that the Permian contribution is minimal. In the study by *Steiner et al.* [1989], samples which were not deemed to be of high enough quality were used only for defining paleomagnetic polarity but not direction. In all but one section, only Triassic samples were considered acceptable and in the other section, Guangyuan, the vast majority are Triassic. In our study, only one site containing the characteristic component is dated as Upper Permian. Considering the relative insignificance of the Upper Permian formations, we conclude that a single pole can be defined which is representative of the Lower Triassic. The Sichuan Basin poles taken

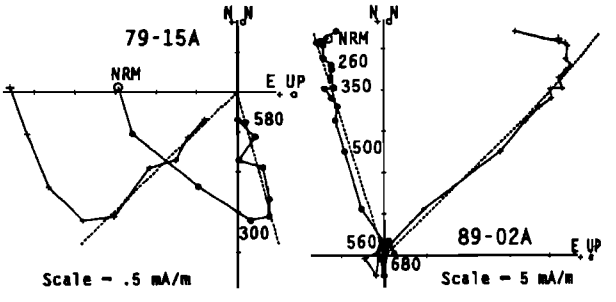


Fig. 2. Typical demagnetization curves in stratigraphic coordinates for samples containing the characteristic component. Demagnetization steps are marked in °C. Points on the vertical (horizontal) projection are marked with circles (plus signs).

Table1

Site	S,D ° , °	n/N	D_G °	I_G °	D_S °	I_S °	k	α_{95} °
79	226,55	13/15	239.2	-10.2	224.8	-16.8	48.8	6.2
89	209,87	6/13	36.2	-17.8	47.4	5.7	33.2	12.3
91*	30,65	9/12	298.1	-25.7	297.8	39.5	103.9	5.1
92	31,51	11/11	27.8	27.5			45.6	6.8
					51.0	19.0	43.0	7.0
93	29,53	10/11	217.7	-15.2	226.1	-1.9	16.3	12.3

Sites with the characteristic component. Site number; Dip direction; n/N: number of specimens measured used in average over the number measured; D, I: mean declination and inclination in (G) geographic and (S) stratigraphic coordinates; k, α_{95}: Fisher precision parameter and 95% confidence limit radius. Starred site shows normal to average great circle.

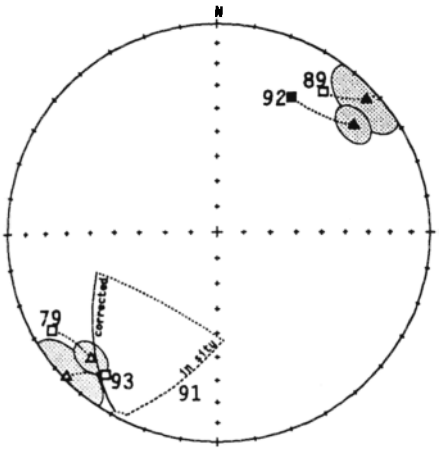


Fig. 3. Site means for the presumed primary component. Lower (upper) hemisphere directions are marked with closed (open) symbols. Squares (triangles) show means in geographic, *in situ* (stratigraphic, tectonically corrected) coordinates. Only a mean constrained great circle could be defined for site 91. The dashed (solid) line marks the *in situ* (tectonically corrected) great circle sector (both the *in situ* and corrected great circle sectors are in the upper hemisphere). The dotted lines show the small circle tectonic corrections about the fold axes.

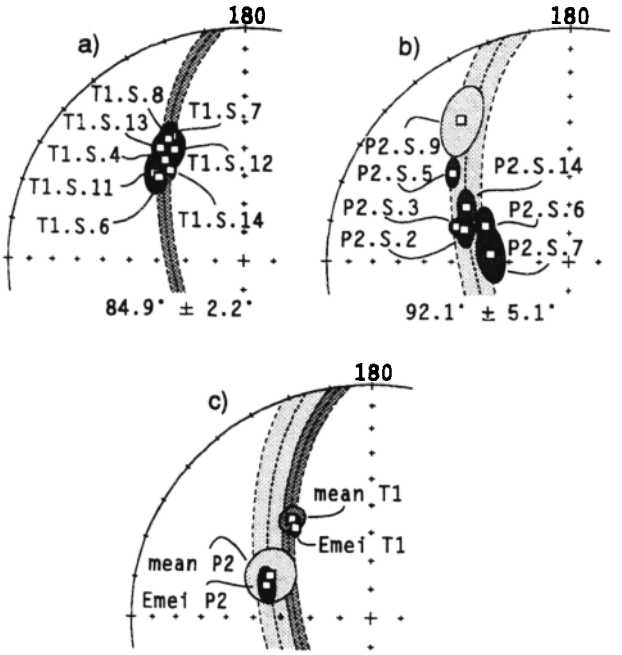


Fig. 4. Compilation of a) Lower Triassic (T1) poles from the Sichuan Basin and b) Uppermost Permian (P2) Emeishan Basalt poles which pass the acceptance criteria of *Enkin et al.* [1992]. Poles are identified as in Figure 1 and in text. Paleolatitudes are calculated from an arbitrary central point (Emei, 29.6°N, 103.4°E, see Figure 1). Pole P2.S.9 was included in the P2 paleolatitude calculation but not in the P2 mean pole. c) The mean T1 Sichuan Basin and P2 Emeishan poles and paleolatitudes are significantly different. The Emei Lower Triassic pole (T1.S.14) is compatible with the mean Lower Triassic pole for the block and the Emeishan Upper Permian Basalt pole (P2.S.2) is compatible with the mean of all Emeishan Basalt poles indicating that the difference between these poles is not only the result of differential strain within the South China Block.

together (Figure 4a) yield a mean pole of 43.8°N, 217.2°E, K=191.9, A_{95}=4.0°, for 8 poles. Note that there are coherent poles from Zhejiang province at the extreme east end of the South China block [*Lin*, 1984, and *Li and Wang*, 1987, *cf. Enkin et al.*, 1992] indicating that the streaking of poles due to local rotations is not severe enough to prevent determination of a single average pole for the entire SCB in the Lower Triassic.

The Emeishan basalt has been studied paleomagnetically over much of its surface (Figure 1). The Red River fault and the Xian Shui He fault both are potential sources for large rotations. In fact, the poles are again streaked, indicating rotations about local vertical axes (Figure 4b). *Enkin* [1990] and *Enkin et al.* [1992] review the available data. Several studies on these rocks do not pass the quite lenient acceptance criteria outlined in *Enkin et al.* [1992] and will not be considered here, but we note that no study in the region gives incompatible results. The accepted poles from the Emeishan Basalt Formation are from Emei P2.S.2 [*McElhinny et al.*, 1981], P2.S.3 [*Zhao and Coe*, 1987] and P2.S.5 [*Liu et al.*, 1985], Zhaojue P2.S.6 [*Liu et al.*, 1985], Yanguan P2.S.7 [*Liu et al.*, 1985], Miyi P2.S.9 [*Zhou et al.*, 1989] and Kunming P2.S.14 [*Fang and Van der Voo*, 1990]. The poles fall on a circle of 92.1° ± 5.1° centred on Emei (method of *McFadden and Reid* [1982]). An average Emeishan Basalt pole has been calculated (neglecting pole P2.S.9 which has the largest uncertainty and is the most rotated) giving 52.7°N, 246.4°E, A_{95}=9.1° (N=6 poles).

The Emeishan remanence is interpreted to be predominately normal [*Zhao and Coe*, 1989], and thus the basalts must have been emplaced after the end of the Kiaman Long Reverse Superchron and date from the Uppermost Permian. The Feixienguan Formation directly overlies the Emeishan Basalt and Xuanwei Sandstone Formations, and is assigned a Lower Triassic age. The maximum possible difference in age according to the timescale of *Harland et al.* [1990] is 10 m.y.

The average paleomagnetic poles of the Feixienguan Formation and the Emeishan Basalts are both reasonably well defined and are significantly different (Figure 4c). Since both the Feixienguan and Emeishan Basalt poles have streaked distributions, we also note that the paleolatitudes (calculated at Emei) are significantly different (7.2° ± 5.6°) at the 95% confidence level. Unfortunately there are no SCB Upper Permian poles taken from sites away from the Emeishan Basalt region, and thus we must consider whether these differences arise from strain within the SCB or imply motion of the entire plate over time (i.e., apparent polar wander).

The Emeishan poles, taken from sites relatively near the Red River Fault, are generally rotated counterclockwise with respect to the Feixienguan poles from the Sichuan Basin. This observation would be compatible with the major left-lateral phase of strike-slip motion on the Red River fault [*Tapponnier et al.*, 1986], but would imply that a folded zone extending some 500 km to the north-east of the fault was sheared by the motion, while the Sichuan Basin, just beyond this zone, remained rigid and unrotated.

Two observations argue against differential motion between the Emeishan region and the Sichuan Basin accounting for the differences between the Emeishan and Feixienguan poles.

First, the Emeishan poles have incompatible paleolatitudes with the Feixienguan poles. Second is the study from the type locality of Emei where samples were taken from a continuous section including Emeishan Basalts and the Feixienguan Formation. We see on Figure 4c that *McElhinny et al.*'s [1981] Emeishan Basalt pole from Emei and *Ma and Zhang*'s [1989] Emei T_{1f} pole are significantly different, but both are indistinguishable (at the 95% confidence level) from the corresponding average poles of all the studies of these two formations. Based on this observation, we are led to accept the hypothesis that the difference is a record of the rotation of the South China block between the Upper Permian and the Lower Triassic.

Conclusion

We have presented Uppermost Permian - Lower Triassic data from the Sichuan Basin. A Mesozoic overprint is found in about half the sites. The component we consider primary is compatible with other poles of the same age around the Sichuan Basin. The main difference between Uppermost Permian Emeishan results and Lower Triassic Sichuan Basin results seems to reflect plate motion of the South China Block rather than to be due only to strain within the block.

Acknowledgements. We thank Z. Zhang and E. Buffetaut for help in the field, and Y. Chen, J. Besse and P. Tapponnier for useful discussions.

References

Enkin, R., *Formation et déformation de l'Asie depuis la fin de l'ère primaire: Les apports de l'étude paléomagnétique des formations secondaires de Chine du Sud*, PhD Thesis, 333 pp., Université de Paris 7, Paris, 1990.

Enkin, R.J., Z. Yang, Y Chen and V. Courtillot, Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present, *J. Geophys. Res., in press*, 1992.

Fang, W., and R. Van der Voo, Paleomagnetism of the Late Permian Emeishan basalt, Yunnan, China, *Eos Trans. AGU*, **71**, 488, 1990.

Harland, W.B., R.L. Armstrong, A. Cox, L.E. Craig, A.G. Smith and D.G. Smith, *A geologic time scale 1989*, Cambridge University Press, 263 pp., Cambridge, 1990.

Heller, F., W. Lowrie, H. Li and J. Wang, Magnetostratigraphy of the Permo-Triassic boundary section at Shangsi (Guangyuan, Sichuan, China), *Earth Planet. Sci. Lett.*, **88**, 348-356, 1988.

Li, H., and Z. Wang, Magnetostratigraphy of the Permo-Triassic boundary at Changxing, Zhejiang province, China, (in Chinese), in *Abstracts for the Third National Symposium on Paleomagnetism*, Guangzhou, China, 1987.

Lin, J., *The Apparent polar wander paths for the North and South China blocks*, Ph.D. thesis, 248 pp, Univ. Calif. Santa Barbara, 1984.

Liu, C., R. Zhu and Z. Jing, Paleomagnetic study of Late Permian Emeishan basalts from Panxi and its neighbouring areas (in Chinese), in *Contributions to Panxi Rift, China, vol. 2*, edited by Y. Zhang, pp. 194-200, Geological Publishing House, Beijing, 1985.

Ma, X., and Z. Zhang, Paleomagnetic study of Permian rocks from Sichuan Emei region and Shanxi Taiyuan region, (in Chinese), *Acta Geophysica Sinica*, **32**, 451-465, 1989.

McElhinny, M.W., B.J.J. Embleton, X.Y. Ma and Z.K. Zhang, Fragmentation of Asia in the Permian, *Nature*, **239**, 212-216, 1981.

McElhinny, M.W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astr. Soc.*, **8**, 338-340, 1964.

McFadden, P.L., and M.W. McElhinny, The combined analysis of remagnetization circle and direct observation in paleomagnetism, *Earth Planet Sci. Lett.*, **87**, 161-172, 1988.

McFadden, P.L., and A.B. Reid, Analysis of palaeomagnetic data, *Geophys. J. R. Astr. Soc.*, **69**, 307-319, 1982.

Opdyke, N., K. Huang, G. Xu and W.Y. Zhang, Paleomagnetic Results from the Triassic of the Yangtze Platform, *J. Geophys. Res.*, **91**, 9553-9568, 1987.

Steiner, M., J. Ogg, Z. Zhang, and S. Sun, The Late Permian/Early Triassic magnetic polarity time scale and plate motions of South China, *J. Geophys. Res.*, **94**, 7343-7363, 1989.

Tapponnier, P., G. Peltzer,and R. Armijo, On the mechanics of the collision between India and Asia, in M.P. Coward and A.C. Ries (Eds.), *Collision tectonics*, Geological Society special publication, **#19**, 115-157, 1986.

Yang, Z., Cheng, Y. and H. Wang, *The Geology of China, Oxford Monographs on Geology and Geophysics*, **#3**, 303 pp., Oxford University Press, 1986.

Zhang, Z.M., J.G. Liou and R.G. Coleman, An outline of plate tectonics of China, *Geol. Soc. Am. Bull.*, **95**, 295-312, 1984

Zhao, X., And R.S. Coe, Paleomagnetic constraints on the collision and rotation of North and South China, *Nature*, **327**, 141-144, 1987.

Zhao, X., And R.S. Coe, Tectonic implications of Permo-Triassic paleomagnetic results from North and South China, in J.W. Hillhouse (Ed.), *Deep structure and past kinematics of accreted terranes*, AGU Geophysics Monograph / IUGG, **5**, 267-283, 1989.

Zhou, Y., L. Lu and B. Zheng, Paleomagnetic polarity of the Permian Emeishan basalt in Sichuan (in Chinese), *Geol. Rev.*, **32**, 465-469, 1986.

V. Courtillot, Ph. Leloup and Z. Yang, Institut de Physique du Globe de Paris, 4 place Jussieu, 75252 Paris Cedex 05, France

R.J. Enkin, Department of Geology and Geophysics, University of Edinburgh, West Mains Road, Edinburgh EH9 3JW, Scotland

L. Xing, J. Zhang, and Z. Zhuang, Chinese Academy of Geological Sciences, Baiwan Zhuang, Beijing, China

(Received August 17, 1992;
Revised September 28, 1992;
Accepted September 29, 1992)